

Disentangling Online Manipulation Strategies from the Perspective of Digital Inequalities

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ABSTRACT

There is a growing concern about the use of manipulative mechanisms in online interfaces and the underlying harm to users' autonomy under the label of 'dark patterns'. The effect of these mechanisms, especially in vulnerable populations, is still under-researched. The present doctoral dissertation aims to understand the conditions under which users are less likely to resist manipulation, following the idea of digital inequalities in manipulative designs. This dissertation expects to contribute in two ways. First, bringing empirical insights about the effectiveness of manipulative mechanisms, especially in low digital skilled populations, and providing enablers that allow them to resist manipulation. Second, providing with guidelines that support practitioners in designing user interfaces for these populations and policymakers in improving the current regulations on manipulative designs.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → **User characteristics**; **Computing literacy**.

KEYWORDS

Dark patterns, deceptive design, manipulative design, digital inequalities

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1 RESEARCH MOTIVATION

There is a growing concern about the use of manipulative mechanisms in online interfaces and the underlying harm to users' autonomy, health, privacy, or economy [25]. Users might for instance be influenced to spend more time on social media [28] to buy more goods on e-commerce platforms [34], or to release more personal data to service providers in exchange for free features [8, 15, 24]. Overall, the concern revolves around users being manipulated into

decisions they initially would not consider making. This risk is amplified by the intricacies between the online domain and everyday activities, e.g., shopping [29], leisure, social media [14], videogames [11], travelling, and streaming services [6], and the widespread use of data to personalise the design of interfaces [3, 8, 17, 26]. The amplitude of these practices, grouped under the label of 'dark patterns', is catching the attention of policymakers, who recently acknowledged this concern based on empirical academic research and investigations around consumer practices [1, 7, 31, 32]. One main problem is the lack of certainty about the specific features that make these manipulative designs so effective. Not only is there a general dearth of empirical research; the particular problem that manipulative mechanisms impact specific users groups more than others has not been researched, either (e.g., socio-economical background, minorities, children). Not enough attention has been paid so far on these vulnerable groups, which are considered worthy of special protection in regulations.

The present doctoral dissertation addresses the impact of manipulative interfaces and mechanisms, especially from the perspective of digital inequalities. It aims at gathering empirical insights and deriving interface design guidelines, that can be worthwhile for policymakers to focus their attention on the most crucial issues in the realm of dark patterns.

2 RELATED WORK

Persuasion in the online domain is defined as trying to influence someone's behaviours transparently by using 'rational' arguments and incentives that the persuadee can resist [4, 5, 36]. In opposition to rational persuasion, manipulation refers to subverting the manipulee's vulnerabilities in a hidden way [35, 36]. To resist this influence, prior work has shown that it is necessary to perceive the try of subversion and have agency over it [2, 3, 23, 27]. Therefore, it is first necessary to understand the users' vulnerabilities towards user interfaces to analyse if they are manipulated or not and, second, test their capability to resist it. Users can be vulnerable online, or less likely to resist manipulation, for different conditions -both personal and external-, and the combination of vulnerabilities makes some users more vulnerable than others [21]. Ecological theories explain, for instance, how users perceive online interfaces differently, building on meso-conditions [18], like education levels, environment, and socio-economic conditions, that can contribute to this position of vulnerability.

In the realm of 'dark patterns', policymakers, like the European Commission or the Federal Trade Commission [7], have attempted to narrow the concept down. Manipulative and deceptive patterns are defined as "design practices that impair autonomy", and scholars have developed attributes and taxonomies [9, 16, 24]. Empirical

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studies have only provided an overview of how some designs can change behaviours. Restricting options in interfaces [30], hiding information, pre-selecting choices or presenting confusing information, or even combining these elements, was shown to be effective at influencing users [20]. Other strategies however (e.g., using emotions, social proof, or scarcity cues [15, 20, 22, 30]) need further analysis to investigate their effects on users. Scholars have demonstrated that users can recognise these practices, but their potential influence or the conditions under which users can resist them are still unclear. Moreover, although the literature on awareness and detection of dark patterns is increasing, the effects these mechanisms have on vulnerable populations -children, the elderly, lower digitally skilled users, and minorities- are under-explored. Bongard-Blanchy et al. [2] and Luguri and Strahilevitz [20] have found manipulative design to be more impactful on users with a lower education level, hinting at a problem that is urgent to address.

Among the potential conditions of vulnerability, it is necessary to analyse the role of these meso-conditions, like socio-digital inequalities. Socio-digital inequalities are *"systematic differences between individuals from different backgrounds in the opportunities and abilities to translate digital engagement into benefits and avoid the harm that might result from engagement with ICTS."* [19](p. 44). In my research, I will consider the digital skills divide to evaluate the differences in the outcomes of users' online engagement, where the outcomes are the differences in resisting manipulation. I will rely on a measure of digital skills as a proxy for digital inequalities and evaluate its role in being manipulated online. Digital skills are well-documented to be impacted by educational levels and economic situation [10, 12, 19, 37–40]. The main hypothesis is that a low level of digital skills may render persuasive elements manipulative.

3 RESEARCH OBJECTIVES

The present research aims to understand the conditions under which users are less likely to resist manipulation, following the idea of digital inequalities in manipulative designs. On the one hand, I aim to explore the design mechanisms of manipulation as an external condition. On the other hand, I aim to investigate the role of digital inequalities in the effects that manipulative designs have on users. I will thus bridge the digital inequalities literature with the HCI domain, trying to understand under which conditions i.e., both external -manipulative designs- and meso -socio-economic conditions- we can consider users vulnerable online and thus less likely to resist manipulation. Knowing the conditions and perceptions of manipulation in this specific population will allow to derive design recommendations for the development of user interfaces.

This dissertation has the following objectives. First, analysing the persuasive design mechanisms that can become manipulative and the actual industrial practices around them (Study 1). Second, understanding how these manipulative design elements are perceived by users with lower digital skills and how they resist them (Study 2). Third, disentangling the factors that can contribute to the resistance to manipulative elements in an experimental setting, analysing a potential divide in the results (Study 3).

Building on the above, I am addressing the following research questions:

- (RQ1) Under which conditions can persuasive mechanisms in user interfaces become manipulative?
- (RQ2) What is the relationship between digital inequalities and resistance to manipulation?

I will explore three dimensions of the interaction between humans and user interfaces: (1) the design mechanisms adopted by practitioners, (2) how users perceive and are impacted by manipulative interfaces, and (3) the outcomes' divides between digitally low vs. high-skilled users.

Through this work, I intend to make the following contributions to the HCI community:

- Providing insights into what design elements, mechanisms and strategies in user interfaces can be problematic in online manipulation. These can take the form of guidelines and heuristics for practitioners, scholars and policymakers that allow them to self-assess when designing for influencing behaviours.
- Providing empirical insights into how underrepresented populations in HCI, e.g., low digital skilled users, use and perceive manipulative technologies. Identifying the enablers that allow resisting manipulation can support ethical practices in design.
- Bridging digital inequalities literature with HCI. By demonstrating a divide in the interaction with manipulative technologies and the causes of this divide, we will provide novel insights into the circumstances under which users are vulnerable online. These findings will serve as the ground for policymakers to improve current regulations on manipulative design, consumer protection and dark patterns.

3.1 Study 1: Manipulation in Practice: What are the Mechanisms of Manipulative Interfaces?

Given the problem of defining dark patterns and what can be considered problematic for users, I decided to disentangle the differences between persuasive and manipulative mechanisms in UX design practices. Persuasive design is inherent to interaction design, but some mechanisms might become manipulative if they subvert vulnerabilities in a hidden way. This distinction was not easy to distinguish in user interfaces, and is sometimes hidden or unknown in common UX practices by designers. Therefore, I conducted a study (study 1) to determine problematic persuasive design mechanisms that might become manipulative UX practices in a simulated scenario where designers are asked to influence users.

I looked at the UX design practices to engage with their mechanisms of influencing users, and put them together with theories of manipulation. It is a qualitative study (n=22) aimed at answering the following research questions:

- (RQ1) How do designers understand different types of influence in design?
- (RQ2) How do designers implement influence mechanisms in design?
- (RQ3) What contextual factors inform designers' decisions when they are asked to influence online?

We elicited initial heuristics to prevent manipulation, informed by the designers' approaches to influence and the literature review. The

Heuristics, therefore, represent design mechanisms that, in combination, can become manipulation. For instance, a non-transparent interface, with a lot of frictional elements that are not supporting users' goals, can be manipulative. In contrast, only a frictional element per se does not necessarily imply a manipulative element.

3.2 Study 2 (envisioned): What Does 'Resistible' Mean for Vulnerable Populations? Perceptions of Manipulation Resistance in Low Digital Skilled Populations

The objective is to understand how low digitally skilled users perceive manipulative techniques, including contextual factors, and if they perceive that manipulative techniques impact their decisions online. It is expected to be a qualitative exploratory study. I will therefore build a qualitative model that gives us an idea of how the design mechanism affects context understanding and resistibility of the incentives as the basis for online manipulation. We aim to make an exploratory analysis of the strategies that might be particularly problematic for specific subpopulations.

- (RQ1) How do low digitally skilled users perceive the use of incentives when manipulated online?
- (RQ2) What are the perceived impacts in their decisions for low digitally skilled users when they are manipulated online?

Study design. This is an exploratory qualitative analysis inspired from [13][33]. Participants will be shown scenarios in which they are presented with familiar situations with manipulative designs -e.g. flight booking, cookie banners, or e-commerce websites. I will conduct a semi-structured interview to explore their perceptions about: (i) how they react against those practices, (ii) what they understand of the mechanism of manipulation, (iii) how they protect themselves from manipulation and (iv) the extent to which they perceive changes in their behaviours caused by those mechanisms. To know the participant's level of digital skills, they will be asked some questions in the screening questionnaire.

3.3 Study 3 (envisioned): What are the Impacts of Digital Skills on Resistance to Manipulation?

The objective of this study is to explore the association between digital inequalities and resistance to manipulation online. Given the strong relationship between socioeconomic inequalities and the digital divides -access, skills and outcomes- that have been widely explored in the literature [12, 19, 37, 39, 40], I will use digital skills as a proxy for digital inequalities. In this regard, the divide in outcomes -or what users can obtain from using technology and being included in society- depends on their online abilities or digital skills. However, there is a strong relationship between how users acquire digital skills and their socio-economic status. In this regard, I will answer the following research questions:

- (RQ1) What is the relationship between digital skills and resistance to manipulation?
- (RQ2) What are the conditions under which users are more likely to resist manipulation?

Therefore, I will explore the relationship between digital skills and resistance to manipulation in two ways. First, I will analyse the

existence of the divide and then the different explainable personal factors that will lead to that divide in resisting manipulation: economic background, education or family education.

Study design. Building on my exploration of the qualitative model that explains the idea of resistance manipulation in low digitally skilled users (See Study 2), I will conduct a large-scale quantitative study. Participants will be presented with a scenario where they will have to spend money on a Fast-Fashion or an NGO website. They will be told that after the task, they will keep the money they do not spend on the product. For example, if the compensation is 20 euros and they pay for a 5 euros product, they will keep the product and only 15 euros instead 20. This is an important aspect to guarantee ecological validity in the scenario. I expect to control the scenario with different incentives according to the qualitative model. I will analyse the differences in behaviour given the level of digital skills, controlling for different socio-economic factors, through difference-in-difference quasi-experimental analysis.

4 NEXT STEPS AND CHALLENGES

The main challenge in finding empirical quantitative evidence to disentangle factors that affect behaviour relates to the validity of the experiments in this specific topic. There are two types of challenges in this regard. The first one is determining the specific design features that work for manipulation. In the online domain, manipulation becomes very context dependent; therefore, it is difficult to attribute in both experimental settings and natural context -e.g. data collection with tracking systems and identifiers online- what are the design elements that affect the user's behaviour. Therefore, I am looking at manipulation mechanisms and strategies to change behaviour. Nonetheless, this challenges operationalising 'resisting the mechanism of influence'.

Setting experimental designs in the online domain with high validity remains problematic and complex, given the difficulty of studying participants in a natural context. Some considerations show a challenge that comes associated with studying online manipulation, given deception becomes necessary in the study design. Therefore, deceiving participants with a high validity in an experimental setting becomes very difficult. Artificial settings present problems given the number of variables that are outside control in a natural context, like the actual willingness or need to acquire a product -in e-commerce settings-; the need for information -in newspaper and news consumption settings- or the actual privacy trade-offs and risks that users make when they have to yield their data online -in the case of cookie banners experiments. When participants are in an artificial setting, they are less likely to consider trade-offs, risks and decisions in the same way they would do in a natural situation. This is one of the common limitations that are presented in the empirical studies in the domain, sometimes leading to limited or inconclusive results. These methodological challenges become specially problematic in the context of low-digital skills users.

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